

JavaScript-based Multiformat Sudoku Puzzle Solver



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Sudoku is a mathematical-logical puzzle where every digit should come uniquely in every row and column without repetition. We generally find Sudoku puzzles in dailies and magazines with various levels of toughness with respect to placement of numbers, number of puzzle

digits and blanks.

Mostly, dailies publish only numerical puzzles with limited matrix length of either 9×9 or 6×6 . In this project, we show that non-numeric entities can also be used to form a Sudoku grid. One can also define the size of the matrix beyond 6×6 and 9×9 . In this project one can use numeric digits from 1-9, and alphabets in upper and lower cases (A-Z, a-z).

The 6×6 matrix with non-numerics is shown in Fig. 1. Likewise, one can have a one-cell grid of 1×1 size to 2401-celled grid of 7×7 size with the above characters.

There are two programs for this project:

sudo_alpha.html and sudo_num.html. The first code is for solving the alphabet-based Sudoku puzzle and the later code is for numeric-based Sudoku puzzle.

In both case, the default size of the matrix is 3×3 , which is a 9×9 matrix having 81-cell grids. User can change the size from 1×1 up to 7×7 as per requirement.

The program

The program is coded in JavaScript and when launched using Internet Explorer, it displays a grid with predefined numbers in each of the cells. The first cell represents the rows, and the second cell represents columns. The rows and columns of the cells define the size of the matrix.

Case 1: When '2' is input in the first cell (column) and '3' is input in the second cell (row), a 6×6 matrix with two columns and three rows is generated, which is colour-coded to differentiate between the distinct cell grids (refer Fig. 2).

Case 2: When '2' is input in the first cell (column) and '2' is input in the second cell (row), a 6×6 matrix with three columns and two rows is generated, which is colour-coded to differentiate between the distinct cell grids (refer Fig. 3).

One more characteristic feature of this program is its artificial intelligence type solving ability that mim-



Fig. 1: 6×6 matrix with non-numerics (unsolved and solved)

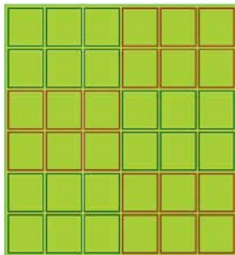


Fig. 2: 2×3 size, 6×6 matrix

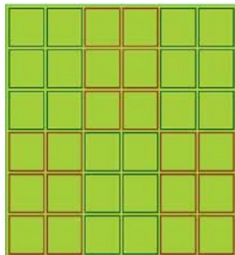


Fig. 3: 3×2 size, 6×6 matrix



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